MILITARY STANDARD

MIL-STD-1325-189
(NAVY)

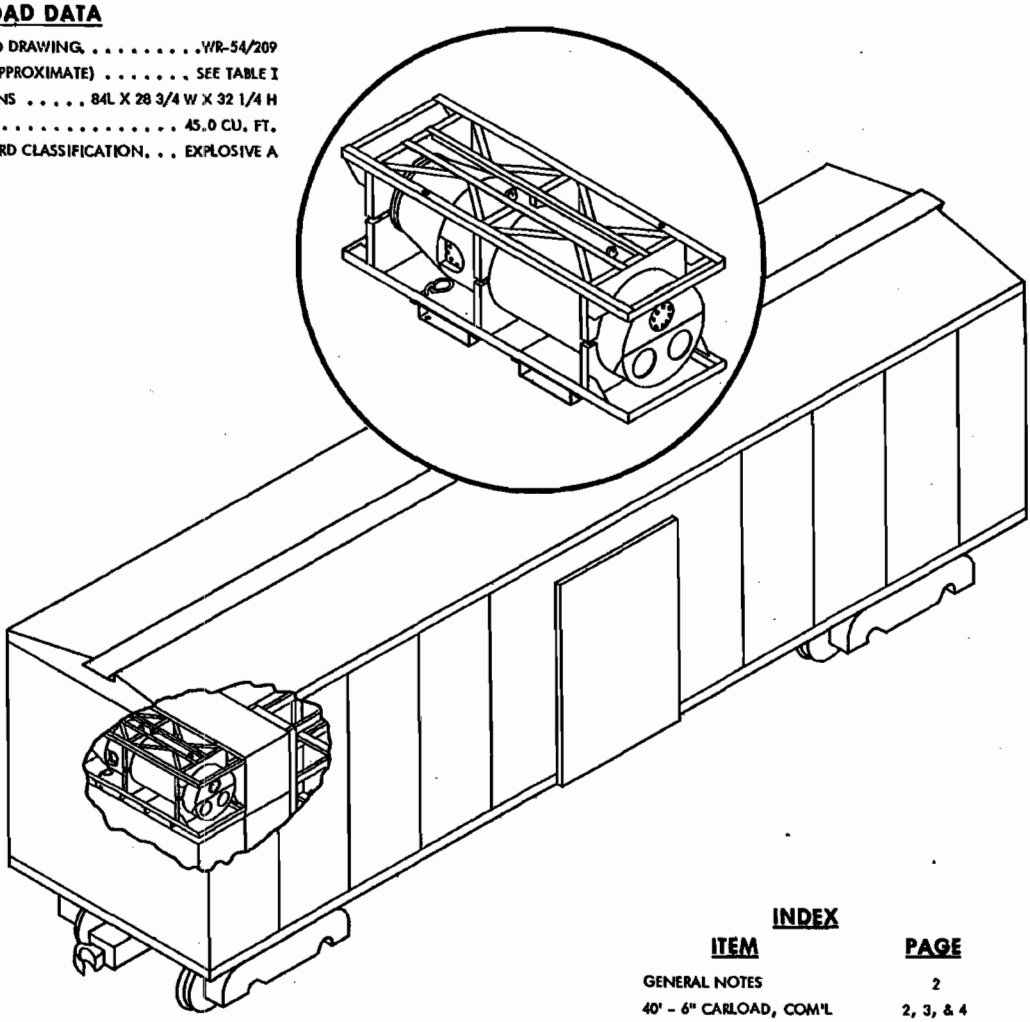
26 SEPTEMBER 1977

RAILCAR LOADING

MINE, UNDERWATER: MK 55 AND MODS, CONFIGURATION "C" AND "D" IN CRATE MK 55 MOD 1 WITH CRATE ADAPTERS MK 112 MOD 0

UNIT LOAD DATA

UNIT LOAD DRAWING. WR-54/209
WEIGHT (APPROXIMATE) . . . . . SEE TABLE I
DIMENSIONS . . . . . 84L X 28 3/4 W X 32 1/4 H
CUBE . . . . . 45.0 CU. FT.
DOT HAZARD CLASSIFICATION. . . EXPLOSIVE A



INDEX

ITEM	PAGE
GENERAL NOTES	2
40' - 6" CARLOAD, COM'L	2, 3, & 4
50' - 6" CARLOAD, COM'L	6, 7, & 8

NOTES:
1. UNLESS OTHERWISE SPECIFIED DIMENSIONS ARE IN INCHES.
2. FOR CROSS REFERENCE TO ASSOCIATED PALLETIZING, TRUCK-LOADING AND CONTAINERLOADING MILITARY STANDARDS, REFER TO INDEX TO STANDARDS MIL-HDBK-236 (NAVY).

FSC 8140

AUTHORIZED AND RELEASED FOR GENERAL USE.

M.S. Bass NAVHC 9/21/77
TECHNICAL DIRECTION AGENT (TDA) DATE

F. Kohl 9/26/77
(AIR) SYSCOM, BY DIRECTION DATE

APPROVED BY BUREAU OF EXPLOSIVES

A. F. Grammuck 9/14/77
SIGNATURE SUPERVISOR, MILITARY AND INTERMODEL SERVICES DATE

ORIGINATOR R F Ebert 9/22/77
DATE

NAVAL WEAPONS HANDLING CENTER
CRANE, INDIANA

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GENERAL NOTES

1. FOR GENERAL INFORMATION CONCERNING ORDERING, INSPECTING, AND PREPARING CARS, AND FOR DUNNAGING MATERIALS, DESIGN, AND INSTALLATION OF DUNNAGE SEE THE GENERAL DOCUMENT MIL-STD-1325 (NAVY) "RAILCAR LOADING OF HAZARDOUS MATERIALS".
2. WHEN PLANNING SHIPMENTS ORDER THE MINIMUM NUMBER OF CARS OF THE CAPACITY REQUIRED FOR THE SHIPMENT.
3. LOADING PLANS SHOWN ARE FOR COMMERCIAL BOXCARS WITH 40 FT. 6 INCHES INSIDE LENGTH, 110 INCHES INSIDE WIDTH, COMMERCIAL BOXCARS WITH 50 FT. 6 INCHES INSIDE LENGTH, 110 INCHES INSIDE WIDTH.
4. THE "LOAD LIMIT" OF A CAR MUST NOT BE EXCEEDED NOR SHOULD THE RAILCAR BE LOADED SO THAT MORE THAN ONE-HALF OF THE "LOAD LIMIT" IS CARRIED BY ONE SET OF TRUCKS.
5. IF END WALLS OF CARS ARE NOT SQUARE THEY MUST BE SQUARED OFF BEFORE STARTING TO LOAD CAR.
6. THE LOAD CONSISTS OF UNDERWATER MINES MK 55 AND MODS IN UNDERWATER MINE CRATE MK 55 MOD 1, WITH CRATE ADAPTERS MK 112 MOD 0 ATTACHED IN ACCORDANCE WITH FLEET ISSUE UNIT LOAD WR-54/209.
7. THE UNIT LOADS ARE HANDLED AND LOADED WITH A SUITABLE FORK LIFT TRUCK.
8. UNLESS OTHERWISE SPECIFIED NAILING SHALL BE IN ACCORDANCE WITH MIL-STD-1325 (NAVY).
9. ALL STRAP JOINTS SHALL BE CRIMP-TYPE JOINTS MADE WITH A CRIMPING TOOL THAT DOES NOT CUT THE SEAL OR STRAP, RATHER THAN A TOOL THAT MAKES A NOTCH-TYPE JOINT. THE STRAPPING MUST BE IN ACCORDANCE WITH NAVSUP NOTICE 4642 DATED 21 APRIL 1971.
10. APPLICABLE MATERIAL SPECIFICATIONS:

    DUNNAGE LUMBER - FED. SPEC MM-L-751

    NAILS - FED. SPEC FF-N-105

    STRAPPING - FED. SPEC QQ-S-781, TYPE I, HEAVY DUTY, FINISH A, B, OR C.

    SEALS - FED. SPEC QQ-S-781, STYLE II, HEAVY DUTY
11. AFTER BLOCKING AND BRACING HAS BEEN INSPECTED ATTACH SHIPPING DOCUMENTS INSIDE THE CAR IN AN ACCESSIBLE AREA, CLOSE AND SEAL BOXCAR DOORS, AND ATTACH APPLICABLE PLACARDS TO THE OUTSIDE OF CAR AS PRESCRIBED IN OP 2165 (VOL 1).

40 FT 6 IN BOXCAR, COMMERCIAL

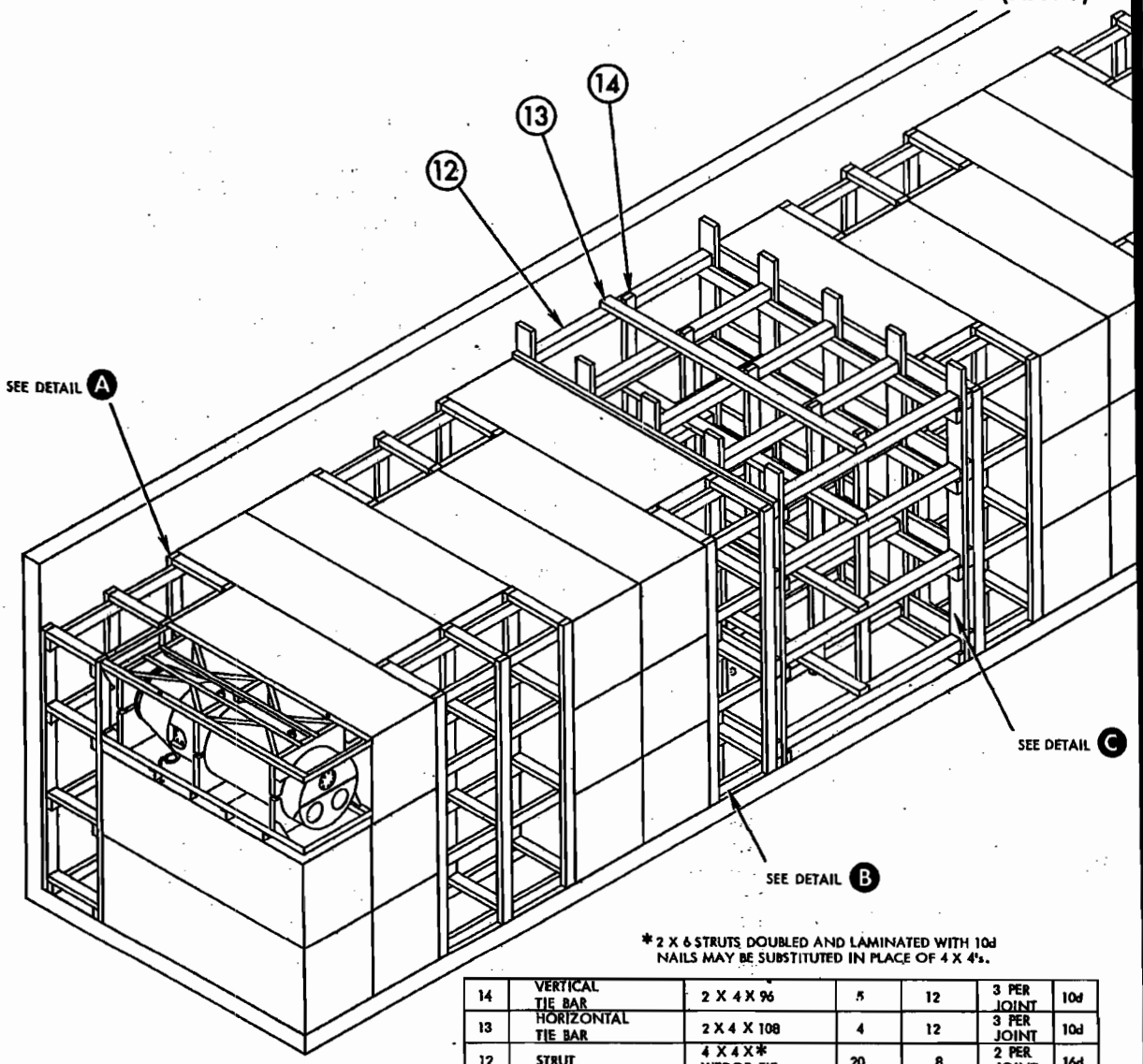
1. THE CARLOAD CONSISTS OF 42 UNIT LOADS WHICH MUST BE LOADED AND DUNNAGED IN ACCORDANCE WITH THIS PROCEDURAL DRAWING.
2. WHEN LESS THAN CARLOAD (LCL) QUANTITIES ARE REQUIRED TO BE SHIPPED IN COMMERCIAL BOXCARS AND A PARTIAL LAYER RESULTS, THE PARTIAL LAYER OF LADING SHALL BE BRACED BY MEANS OF END BRACING AND/OR PARTIAL LAYER BRACING CONSTRUCTED IN ACCORDANCE WITH MIL-STD-1325-100 (NAVY) OR PARTIAL UPPER LAYER BRACING IN ACCORDANCE WITH MIL-STD-1325-102 (NAVY). SELECT THE TYPE OF BRACE TO COMPLY WITH THE WEIGHT OF THE UNITS TO BE RETAINED. THE WEIGHT DISTRIBUTION REQUIREMENTS OF THE AAR MUST BE COMPLIED WITH, SEE MIL-STD-1325 (NAVY).
3. THE LOADS AS SHOWN ARE BASED ON CARS WHICH HAVE 6 FT. WIDE DOORWAY OPENINGS AND ARE EQUIPPED WITH CONVENTIONAL SLIDING TYPE DOORS. THE DEPICTED PROCEDURES AND METHODS OF BLOCKING ARE APPLICABLE TO BOXCARS EQUIPPED WITH CONVENTIONAL SLIDING TYPE DOORS OTHER THAN 6 FT. WIDE.

TABLE I

DODIC/ NALC NO.	MINE MK 55 MOD	CONFIG-URATION	UNIT LOAD WT. (LBS. APPROX.)	CARLOAD WT. (LBS. APPROX.)*
U876	1	C	2305	99174
U877	1	D	2313	99510
U888	2	C	2379	102282
U889	2	D	2371	101946
U900	3	C	2389	102702
U901	3	D	2381	102366
U912	4	C	2318	99720
U913	4	D	2310	99384
U924	5	C	2386	102576
U925	5	D	2382	102408
U930	6	C	2397	102968
U931	6	D	2382	102408
U936	7	C	2168	93420
U937	7	D	2558	109800
U942	8	C	2411	103626
U943	8	D	2558	109800

*ESTIMATED WEIGHT OF DUNNAGE (2364 LBS.) INCLUDED IN CARLOAD WEIGHT.

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* 2 X 6 STRUTS DOUBLED AND LAMINATED WITH 10d NAILS MAY BE SUBSTITUTED IN PLACE OF 4 X 4's.

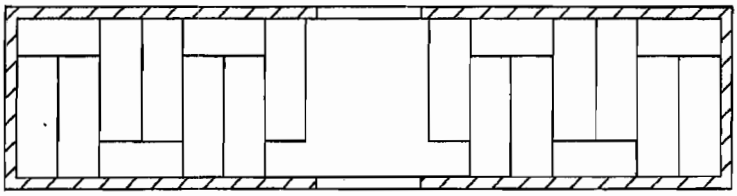
PIECE NO.	DESCRIPTION	SIZE	NO. OF PIECES REQ'D	NAIL TO	NAILS NUMBER	NAILS SIZE
LIST OF MATERIALS & NAILING DATA						
14	VERTICAL TIE BAR	2 X 4 X 96	5	12	3 PER JOINT	10d
13	HORIZONTAL TIE BAR	2 X 4 X 108	4	12	3 PER JOINT	10d
12	STRUT	4 X 4 X* WEDGE FIT	20	8	2 PER JOINT	16d
11	HOLD DOWN	2 X 3 X TO SUIT	4	10	3	10d
10	CENTER GATE FILLER	2 X 4 X CAR WIDTH - 1	2	8	3 PER JOINT	10d
9	CENTER GATE STRUT CLEAT	2 X 4 X CAR WIDTH - 1	8	8	3 PER JOINT	10d
8	CENTER GATE VERTICAL	2 X 6 X 108	10	SEE 7, 9	-	-
7	CENTER GATE HORIZONTAL	2 X 6 X CAR WIDTH - 1	8	8	3 PER JOINT	10d
6	CROSS BRACE	2 X 4 X TO SUIT	16	5	2 PER JOINT	10d
5	SWAY BRACE HORIZONTAL	2 X 4 X 28 3/4	16	4	3 PER JOINT	10d
4	SWAY BRACE VERTICAL	2 X 4 X 97	8	SEE 5	-	-
3	CROSS BRACE	2 X 4 X TO SUIT	72	2	2 PER JOINT	10d
2	SWAY BRACE HORIZONTAL	2 X 4 X 57 1/2	48	1	3 PER JOINT	10d
1	SWAY BRACE VERTICAL	2 X 4 X 97	36	SEE 2	-	-

CARLOAD DATA

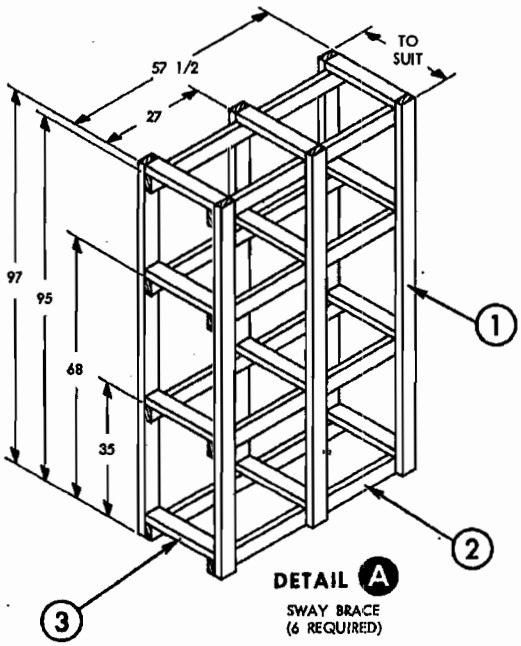
NUMBER OF UNIT LOADS 42

CARLOAD WEIGHT. . . . . SEE TABLE I

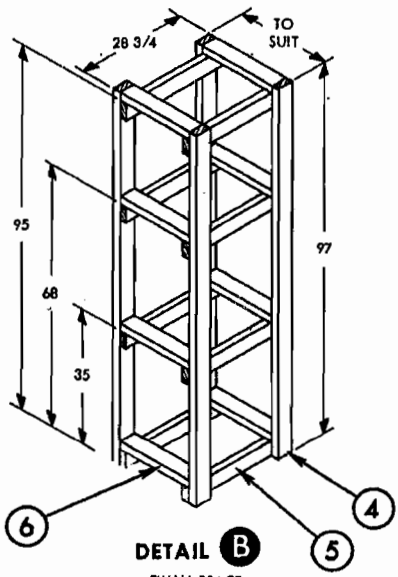
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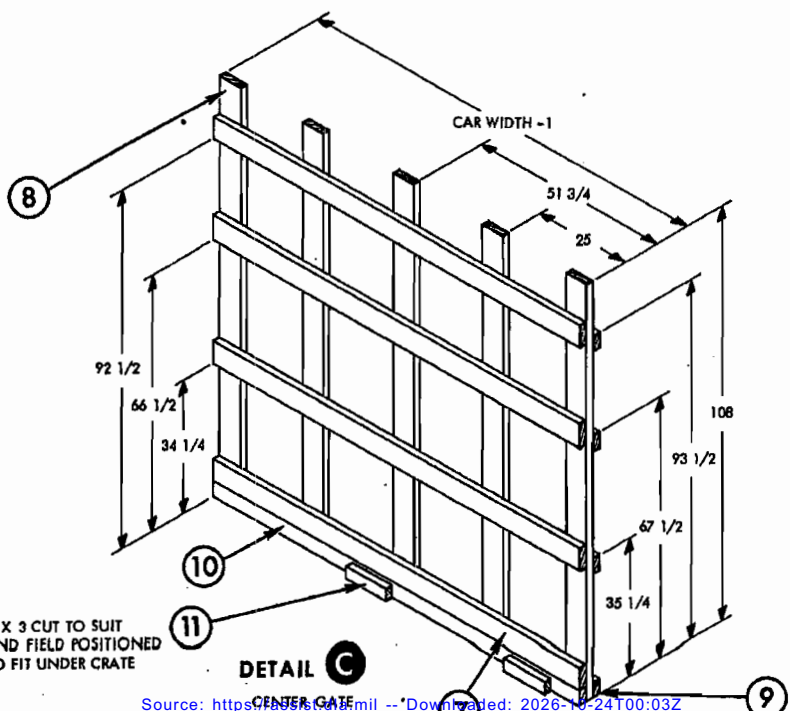
LOADING PLAN



DETAIL A
SWAY BRACE
(6 REQUIRED)



DETAIL B
SWAY BRACE
(2 REQUIRED)



DETAIL C
CENTER GATE
(2 REQUIRED)

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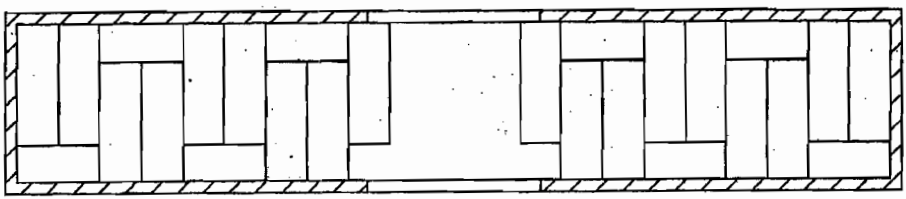
50 FT 6 IN BOXCAR, COMMERCIAL

1. THE CARLOAD CONSISTS OF 54 UNIT LOADS WHICH MUST BE LOADED AND DUNNAGED IN ACCORDANCE WITH THIS PROCEDURAL DRAWING.
2. WHEN LESS THAN CARLOAD (LCL) QUANTITIES ARE REQUIRED TO BE SHIPPED IN COMMERCIAL BOXCARS AND A PARTIAL LAYER RESULTS, THE PARTIAL LAYER OF LADING SHALL BE BRACED BY MEANS OF END BRACING AND/OR PARTIAL LAYER BRACING CONSTRUCTED IN ACCORDANCE WITH MIL-STD-1325-100 (NAVY) OR PARTIAL UPPER LAYER BRACING IN ACCORDANCE WITH MIL-STD-1325-102 (NAVY). SELECT THE TYPE OF BRACE TO COMPLY WITH THE WEIGHT OF THE UNITS TO BE RETAINED. THE WEIGHT DISTRIBUTION REQUIREMENTS OF THE AAR MUST BE COMPLIED WITH, SEE MIL-STD-1325 (NAVY).
3. THE LOADS AS SHOWN ARE BASED ON CARS WHICH HAVE 10 FT. WIDE DOORWAY OPENINGS AND ARE EQUIPPED WITH CONVENTIONAL SLIDING TYPE DOORS. THE DEPICTED PROCEDURES AND METHODS OF BLOCKING ARE APPLICABLE TO BOXCARS EQUIPPED WITH CONVENTIONAL SLIDING TYPE DOORS OTHER THAN 10 FT. WIDE.
4. THE DEPICTED PROCEDURES AND METHODS OF BLOCKING ARE ALSO APPLICABLE TO BOXCARS EQUIPPED WITH PLUG DOORS PROVIDING STACKS WITH MORE THAN HALF OF THE UNIT LOAD IN THE DOORWAY AREA ARE UNITIZED WITH TWO LATERALLY APPLIED 1 1/4" STEEL STRAPS PER STACK EACH TENSIONED AND SEALED WITH TWO DOUBLE CRIMPED SEALS OR, AS AN ALTERNATE, EACH PLUG DOOR MAY BE SPANNED WITH FOUR 2 X 6 MEMBERS LOCATED AT THE SAME HEIGHTS AS THE CENTER GATE HORIZONTALS AND NAILED TO THE SIDE WALLS WITH FIVE 10d NAILS EACH END. DUNNAGE MATERIAL MUST NOT BE NAILED TO ANY PLUG DOOR, WHETHER MAIN OR AUXILIARY, EXCEPT WHEN THE CAR HAS A COMBINATION OF A CONVENTIONAL SLIDING TYPE DOOR AND A PLUG TYPE DOOR AND AN ADEQUATE NAILING STRIP IS PROVIDED ON THE PLUG TYPE DOOR. IF LUMBER OF SUFFICIENT LENGTH TO SPAN PLUG DOORS IS NOT AVAILABLE RANDOM LENGTH MATERIAL, DOUBLED AND SPLICED BUT WITH JOINTS OF SPLICES OFFSET, MAY BE USED. DIMENSIONAL LUMBER DOORWAY PROTECTION, PIECES 15 AND 16, IS NOT REQUIRED WHEN PLUG DOOR EQUIPPED BOXCARS ARE USED EXCEPT WHEN CAR HAS A COMBINATION OF PLUG DOOR AND CONVENTIONAL SLIDING DOOR, THEN DIMENSIONAL LUMBER DOORWAY PROTECTION IS REQUIRED FOR THE CONVENTIONAL DOOR. PLUG DOORS MUST BE SECURLY CLOSED AND WIRED TOGETHER WITH A STRONG FLEXIBLE STEEL WIRE INSERTED THROUGH THE HOLES IN THE DOOR LATCH ASSEMBLY ONE OR MORE TIMES AND THE WIRE ENDS TWISTED TOGETHER.

TABLE II

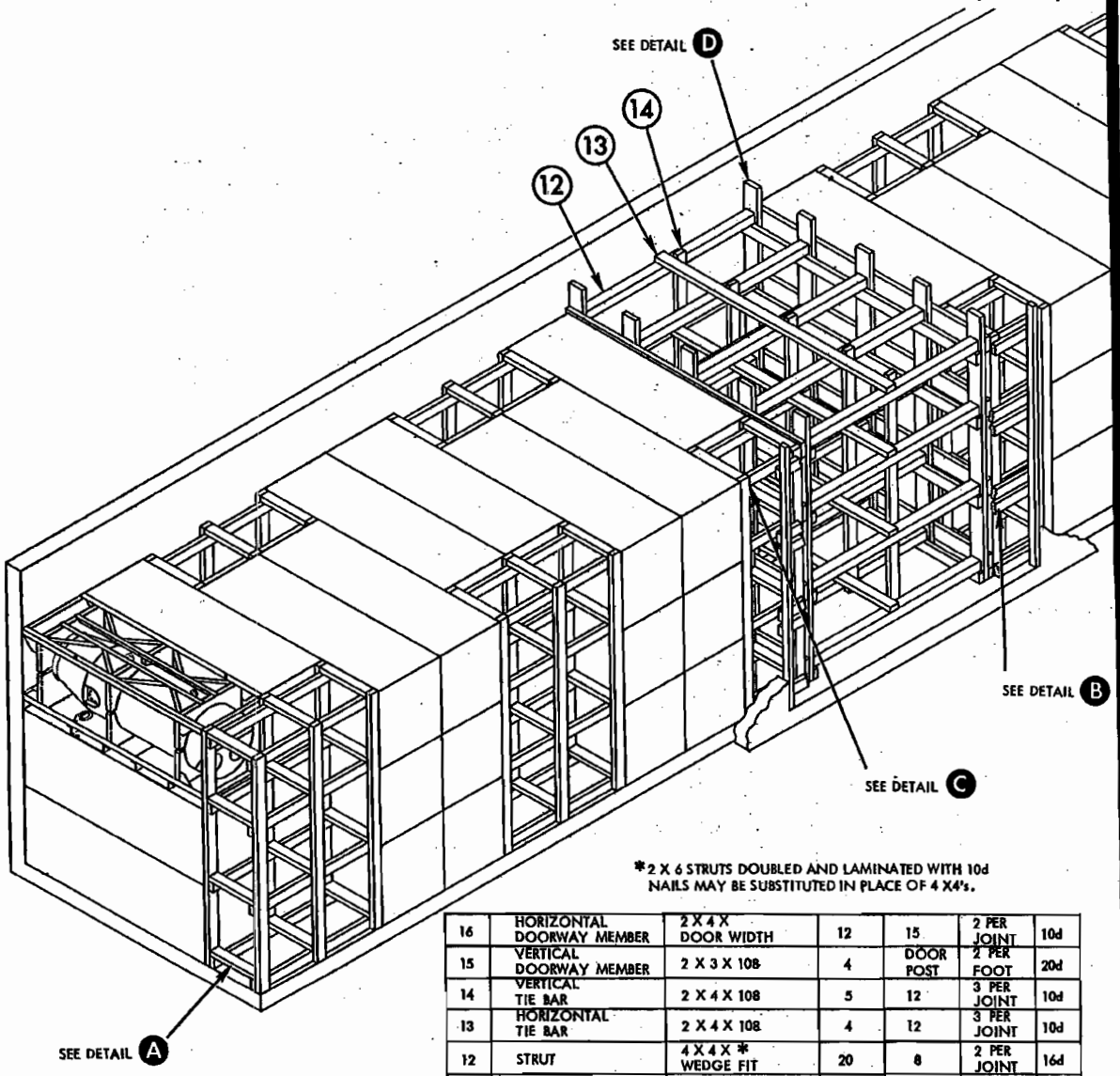
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U877	1	D	2313	127542
U888	2	C	2379	131106
U889	2	D	2371	130674
U900	3	C	2389	131646
U901	3	D	2381	131214
U912	4	C	2318	127812
U913	4	D	2310	127380
U924	5	C	2386	131484
U925	5	D	2382	131268
U930	6	C	2397	132078
U931	6	D	2382	131268
U936	7	C	2168	119712
U937	7	D	2558	140772
U942	8	C	2411	132834
U943	8	D	2558	140772

*ESTIMATED WEIGHT OF DUNNAGE (2640 LBS.) INCLUDED IN CARLOAD WEIGHT



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*2 X 6 STRUTS DOUBLED AND LAMINATED WITH 10d NAILS MAY BE SUBSTITUTED IN PLACE OF 4 X4's.

PIECE NO.	DESCRIPTION	SIZE	NO. OF PIECES REQ'D	NAIL TO	NUMBER NAILS	SIZE
16	HORIZONTAL DOORWAY MEMBER	2 X 4 X DOOR WIDTH	12	15	2 PER JOINT	10d
15	VERTICAL DOORWAY MEMBER	2 X 3 X 108	4	DOOR POST	2 PER FOOT	20d
14	VERTICAL TIE BAR	2 X 4 X 108	5	12	3 PER JOINT	10d
13	HORIZONTAL TIE BAR	2 X 4 X 108	4	12	3 PER JOINT	10d
12	STRUT	4 X 4 X * WEDGE FIT	20	8	2 PER JOINT	16d
11	HOLD DOWN	2 X 3 X TO SUIT	4	10	3	10d
10	CENTER GATE FILLER	2 X 4 X CAR WIDTH - 1	2	8	3 PER JOINT	10d
9	CENTER GATE STRUT CLEAT	2 X 4 X CAR WIDTH - 1	8	8	3 PER JOINT	10d
8	CENTER GATE VERTICAL	2 X 6 X 108	10	SEE 7, 9	-	-
7	CENTER GATE HORIZONTAL	2 X 6 X CAR WIDTH - 1	8	8	3 PER JOINT	10d
6	CROSS BRACE	2 X 4 X TO SUIT	16	5	2 PER JOINT	10d
5	SWAY BRACE HORIZONTAL	2 X 4 X 28 3/4	16	4	3 PER JOINT	10d
4	SWAY BRACE VERTICAL	2 X 4 X 97	8	SEE 5	-	-
3	CROSS BRACE	2 X 4 X TO SUIT	96	2	2 PER JOINT	10d
2	SWAY BRACE HORIZONTAL	2 X 4 X 57 1/2	64	1	3 PER JOINT	10d
1	SWAY BRACE VERTICAL	2 X 4 X 97	48	SEE 2	-	-

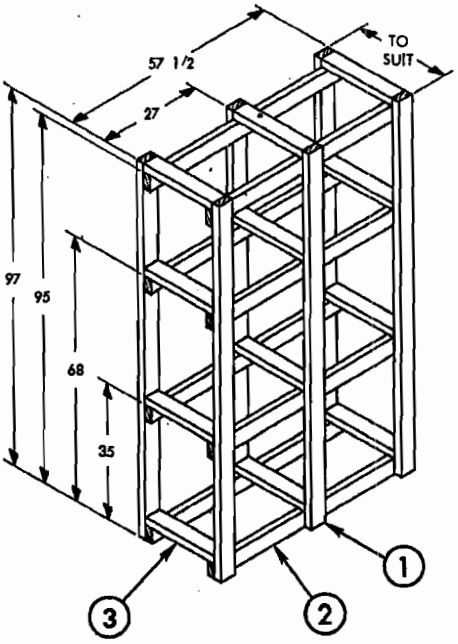
LIST OF MATERIALS & NAILING DATA

CARLOAD DATA

NUMBER OF UNIT LOADS 54

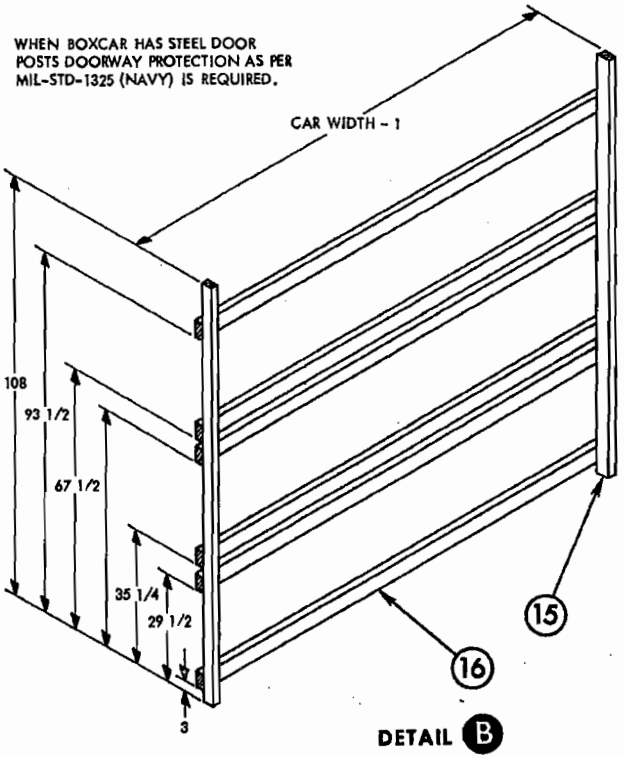
CARLOAD WEIGHT. . . . . .SEE TABLE II

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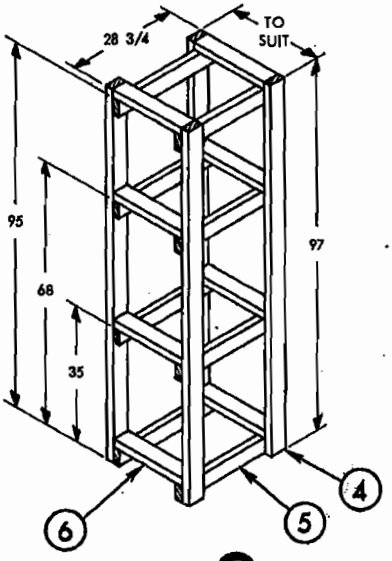


DETAIL A
SWAY BRACE
(6 REQUIRED)

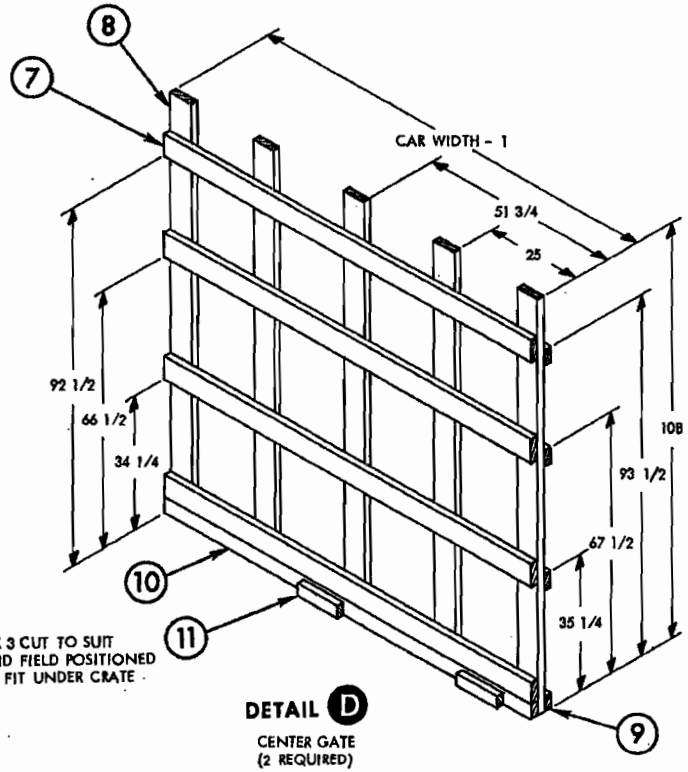
WHEN BOXCAR HAS STEEL DOOR POSTS DOORWAY PROTECTION AS PER MIL-STD-1325 (NAVY) IS REQUIRED.



DETAIL B
DOORWAY PROTECTION
(2 REQUIRED)



DETAIL C
SWAY BRACE
(2 REQUIRED)



2 X 3 CUT TO SUIT AND FIELD POSITIONED TO FIT UNDER CRATE

DETAIL D
CENTER GATE
(2 REQUIRED)

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